Course 1, Module 4 Activity: Cybersecurity Tools

Google Cybersecurity Professional Certificate · Foundations of Cybersecurity

Date: 13 October 2026 · Expected time: 60–90 minutes · Submit one document with all your answers, labelled by part and question number.

Scenario: Your First Monday as a Security Analyst

You just joined BrightBake Online, a small e-commerce bakery that sells cakes through its website, as a junior security analyst. On Monday morning your manager says: "Our SIEM fired some alerts over the weekend. Walk through them using our tools and tell me what happened."

This activity covers every topic in Course 1, Module 4 (Cybersecurity tools): logs, SIEM tools and dashboards, playbooks, network protocol analyzers, the core analyst tools (Linux, SQL, Python), and your cybersecurity portfolio.

How to complete it

- Work through Parts 1 to 6 in order. Each part builds on the same incident.
- Write your answers in your own words. Short, clear answers are better than long copied definitions.
- Expected time: 60 to 90 minutes.
- Submit one document with all your answers, labelled by part and question number.

All IP addresses and company names below are fictional and used for practice only.

Part 1: Read the Logs

A log is a record of events that happen inside an organization's systems. Your SIEM collected the three log extracts below from BrightBake's systems on Sunday night.

Log A

```
2026-10-12 02:14:07  DENY  TCP  203.0.113.45:51544 -> 10.0.0.12:22
2026-10-12 02:14:09  DENY  TCP  203.0.113.45:51545 -> 10.0.0.12:22
2026-10-12 02:14:11  DENY  TCP  203.0.113.45:51546 -> 10.0.0.12:3389
2026-10-12 02:19:58  ALLOW TCP  203.0.113.45:51600 -> 10.0.0.12:22
```

Log B

```
2026-10-12 02:20:31  auth-server  sshd: Failed password for admin from 203.0.113.45
2026-10-12 02:20:36  auth-server  sshd: Failed password for admin from 203.0.113.45
2026-10-12 02:20:41  auth-server  sshd: Failed password for admin from 203.0.113.45
2026-10-12 02:20:47  auth-server  sshd: Failed password for admin from 203.0.113.45
2026-10-12 02:21:02  auth-server  sshd: Accepted password for admin from 203.0.113.45
```

Log C

```
2026-10-12 02:25:44  src=10.0.0.12  dst=198.51.100.7:443  bytes_out=48,213,992
2026-10-12 02:26:10  src=10.0.0.12  dst=198.51.100.7:21   bytes_out=12,004
```

1. Which type of log is each extract: firewall log, network log, or server log? Give one clue from the log for each answer.
2. Put the events from all three logs into one timeline. In 3 to 4 sentences, describe what you think happened.
3. Which IP address belongs to the attacker, and which belongs to BrightBake's server?
4. Which element of the CIA triad (confidentiality, integrity, availability) is most at risk here? Explain why.
5. Why are accurate timestamps so important when analysts compare logs from different sources?

Part 2: SIEM Tools and Dashboards

BrightBake's SIEM collects logs from many sources and shows them on one dashboard. This is what the analyst view looked like on Monday morning.

Alert name	Severity	Count (last 24 h)	Source
Multiple failed SSH logins, then success	High	1	auth-server
Large outbound data transfer	High	1	network sensor
Blocked port scan	Medium	3	firewall
Expired TLS certificate warning	Low	1	web-server
New user added to admin group	Medium	1	auth-server

1. In your own words, what is a SIEM tool and what does SIEM stand for?
2. How does a dashboard help an analyst who receives thousands of log entries a day?
3. Rank the five alerts above from first to last in the order you would investigate them. Explain your top two choices.
4. The "New user added to admin group" alert was not in Part 1. How could it connect to the same incident?
5. Match each SIEM tool to its type: Splunk Enterprise, Splunk Cloud, Google SecOps (Chronicle). Types: self-hosted, cloud-hosted, cloud-native.
6. Which SIEM hosting type (self-hosted, cloud-hosted, or hybrid) fits each company below? Give one reason each.
   - **a.** A bank that must keep all data on its own servers by law.
   - **b.** A new startup with no physical servers and a small IT team.
   - **c.** BrightBake, which keeps its order database in its office but is moving its website to the cloud.

Part 3: Follow the Playbook

A playbook is a manual that gives details about the operational actions to take in a specific situation. BrightBake has a playbook called "Suspicious Login and Possible Data Exfiltration." Its steps got shuffled when someone copied it.

- A. Block the attacker's IP address at the firewall and disable the compromised admin account.
- B. Receive and acknowledge the alert from the SIEM.
- C. Write an incident report and note lessons learned.
- D. Check the related logs to confirm the alert is real and not a false positive.
- E. Escalate to the incident response team and inform your manager.
- F. Reset all admin passwords, turn on multi-factor authentication (MFA), and restore normal operations.

1. Put steps A to F in the correct order. Write the letters only, for example: B, D, ...
2. Why should an analyst follow a playbook instead of improvising during an incident? Give two reasons.
3. At which step would a SIEM tool help most, and at which step would a network protocol analyzer help most?
4. Playbooks are living documents. After this incident, suggest one new step BrightBake should add to this playbook.

Part 4: Network Protocol Analyzer

A network protocol analyzer (also called a packet sniffer) captures and analyzes data traffic on a network. Your manager ran tcpdump on BrightBake's server and saved this short capture.

tcpdump capture

```
02:25:44.101 IP 10.0.0.12.49822 > 198.51.100.7.443: Flags [P.], length 1448
02:25:44.102 IP 10.0.0.12.49822 > 198.51.100.7.443: Flags [P.], length 1448
02:25:44.103 IP 10.0.0.12.49822 > 198.51.100.7.443: Flags [P.], length 1448
02:26:10.004 IP 10.0.0.12.49830 > 198.51.100.7.21: Flags [S], length 0
```

1. For the first line, write the source IP, source port, destination IP, and destination port.
2. Which entry in Part 1 does this capture confirm? Explain how.
3. Port 443 is HTTPS and port 21 is FTP. Why might an attacker's use of FTP be useful to an analyst trying to see what was stolen?
4. tcpdump and Wireshark are both network protocol analyzers. What is the main difference between how you use them?
5. Attackers can also use packet sniffers. Describe one way an attacker could misuse one.
6. Ethics check: before capturing network traffic at a real job, what must an analyst have, and why?

Part 5: Choose the Right Tool (Linux, SQL, Python)

While cleaning up, your manager gives you six tasks. For each one, write which tool fits best, Linux, SQL, or Python, and one sentence explaining why.

#	Task	Your tool	Why
1	Search a database of 50,000 login records for every attempt from 203.0.113.45		
2	Open the server's command line and find the authentication log file		
3	Write a script that checks new IP addresses against a blocklist every hour, automatically		
4	Filter the employee table to list everyone who logged in between midnight and 6 a.m.		
5	Check and change the file permissions on the server so only admins can read the logs		
6	Automate an email to the security team whenever a High severity alert appears		

Bonus (optional): Try writing the SQL query for Task 1. Assume the table is called log_in_attempts and the column is called ip_address.

Part 6: Add It to Your Portfolio

A cybersecurity portfolio shows employers what you can actually do. Turn this activity into your first portfolio piece.

1. Write a 150 to 200 word incident summary for a non-technical manager. Cover what happened, which tools you used, and what BrightBake should do next. Avoid jargon.
2. List the tools from this activity you would add to your resume's skills section, and next to each, one thing you now understand about it.
3. Choose where you will keep your portfolio (for example a Google Drive folder, GitHub, or a personal website) and give one reason for your choice.

End of activity. Good luck, analyst — Monday is yours.